

Database Management System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What is an E-R Diagram? Explain various symbols used in the E-R Diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is Transaction? State and explain the properties of Transaction.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. From the relation given below, answer the following questions.
EMPLOYEE (Ssn, Fname, Minit, Name, Bdate, Address, Sex, Salary, Super_ssn, Dno)
DEPARTMENT (Dname, Dnumber, Mgr_ssn, Mgr_start_date)
PROJECT (Pname, Pnumber, Plocation, Dnum)
DEPENDENT (Essn, Dependent_name, Sex, Bdate, Relationship)
DEPT_LOCATIONS (Dnumber, Dlocation)
a) Retrieve all the attributes of employees who work for the 'Computer' department
b) Retrieve department no 5 using SQL
b) For each Department, retrieve the department number, the number of employees in department, and their average salary using SQL
c) For each project in which more than two employees work, retrieve the project number, name, and the number of employees who work on that project using SQL
d) Select the EMPLOYEE tuples whose department is 10, or those whose salary is greater than Rs 50,000 using Relational Algebra.
e) Retrieve the names of employees who have no dependents using Relational Algebra

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What are different types of Database Users? Explain in Brief.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is Primary key and Foreign Key? Why is this required in the DBMS?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is Integrity constraint? What are the three main categories of Integrity constraints?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is Functional Dependency? Explain 3NF with Examples.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What is Schedule? Explain Serializability and Conflict Schedule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is starving in DBMS? Explain with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. During ER-to-Relational Mapping, explain with example to map 1: N and N:M relationship into a relation?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is NoSQL? What are the types of NoSQL? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Explain Data Models, Schemas, and Instances.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.